Seth Ricks ECEN380 PSOB: 3.55(b), 3.56(d), 3.57(b), 3.57(d), 3.58(b), 3.60(c), MATLAB 3.112

3.55(b) Use the equation describing the FT representation to determine the time-domain signals corresponding to the following FTs: 1/6

b) $X(j\omega) = e^{-2\omega}u(\omega)$

FT: $x(t) = \frac{1}{2\pi}\int_{-\infty}^{\infty} X(j\omega)e^{j\omega t}\,d\omega$

$X(j\omega) = e^{-2\omega}u(\omega) = e^{-2\omega}, \quad \omega > 0$

$x(t) = \frac{1}{2\pi}\int_0^{\infty} e^{-2\omega}e^{j\omega t}\,d\omega = \frac{1}{2\pi}\int_0^{\infty} e^{(j\omega t - 2\omega)}\,d\omega = \frac{1}{2\pi}\int_0^{\infty} e^{\omega(jt-2)}\,d\omega$

$= \frac{1}{2\pi}\cdot\frac{e^{\omega(jt-2)}}{jt-2}\Big|_0^{\infty} = \frac{1}{2\pi(jt-2)}\cdot e^{\omega(jt-2)}\Big|_0^{\infty} = \frac{1}{2\pi(jt-2)}\cdot e^{-\omega(2-jt)}\Big|_0^{\infty}$

$= \frac{1}{2\pi(jt-2)}\cdot\left[e^{-\infty} - e^{0}\right] = \frac{1}{2\pi(jt-2)}\cdot\left[0 - 1\right]$

$$\boxed{x(t) = \frac{-1}{2\pi(jt-2)}}$$

3.56(d) Determine the appropriate Fourier representation of the following time-domain signals, using the defining equations:

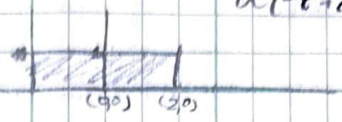
2/6

(d) $x(t) = e^{(1+t)}u(-t+2)$

Cont., Time domain → FT

$u(-t+2) = u(-1(t-2))$

FT: $X(j\omega) = \int_{-\infty}^{\infty} x(t)e^{-j\omega t}\,dt$



$x(t) = e^{(1+t)}u(-t+2) = e^{(1+t)},\ t \le 2$

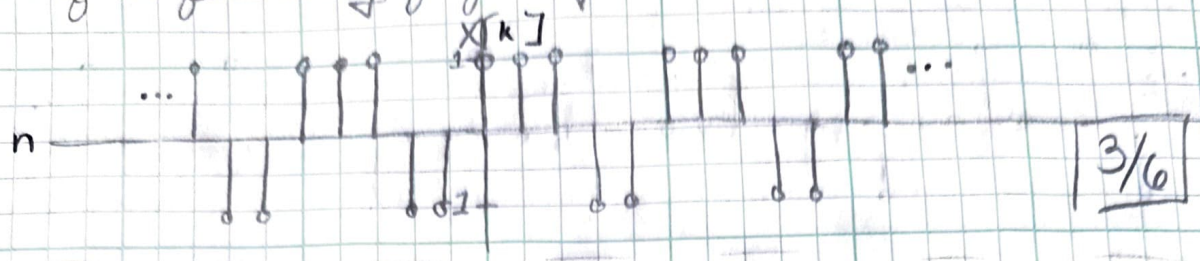
$X(j\omega) = \int_{-\infty}^{-2} e^{(1+t)}e^{-j\omega t}\,dt = \int_{-\infty}^{-2} e^{(1+t-j\omega t)}\,dt = \int_{-\infty}^{-2} e^{(1+t(1-j\omega))}\,dt$

$= \int_{-\infty}^{-2} e^{1}e^{t}e^{(1-j\omega)}\,dt = e^{(1-j\omega)}\int_{-\infty}^{-2} e^{t}\,dt = e^{(1-j\omega)}\cdot e^{t}\Big|_{-\infty}^{-2}$

$= e^{(1-j\omega)}\cdot[e^{-2} - \cancel{e^{-\infty}}^{\,0}] = e^{(1-j\omega)}e^{-2} = e^{(1-j\omega-2)} = e^{(-1-j\omega)}$

$\boxed{X(j\omega) = e^{(-1-j\omega)}}$

3.57(b) Determine the time-domain signal corresponding to each of the following frequency-domain representations:



3/6

Frequency, Discrete, periodic → DTFS

$$x[n] = \sum_{k=0}^{N-1} X[k] e^{jk\Omega_0 n} \; ; \; X[k] = \frac{1}{N}\sum_{n=0}^{N-1} x[n] e^{-jk\Omega_0 n}$$

$$N = 5 \rightarrow \Omega_0 = \frac{2\pi}{5}$$

$$x[n] = \sum_{k=0}^{4} X[k] e^{jk\frac{2\pi}{5}n} \qquad X[k] = \begin{cases} 1, & n = 0,1,2 \\ -1, & n = 3,4 \end{cases}$$

$$x[n] = x[0]e^{j(0)\frac{2\pi}{5}n} + x[1]e^{j(1)\frac{2\pi}{5}n} + x[2]e^{j(2)\frac{2\pi}{5}n} + x[3]e^{j(3)\frac{2\pi}{5}n} + x[4]e^{j(4)\frac{2\pi}{5}n}$$

$$x[n] = 1 + e^{j\frac{2\pi}{5}n} + e^{j\frac{4\pi}{5}n} + -e^{j\frac{6\pi}{5}n} + -e^{j\frac{8\pi}{5}n}$$

$$x[n] = 1 + e^{j\Omega_0 n} + e^{j2\Omega_0 n} + \underbrace{-e^{j3\Omega_0 n}}_{=-e^{j(3-5)\Omega_0 n}} + \underbrace{-e^{j4\Omega_0 n}}_{=-e^{j(4-5)\Omega_0 n}}$$

(w/ period $N=5$)

$$x[n] = 1 + e^{j\Omega_0 n} + e^{j2\Omega_0 n} + -e^{-j2\Omega_0 n} - e^{-j\Omega_0 n}$$

$$x[n] = 1 + j2\left(\frac{e^{j\Omega_0 n} - e^{-j\Omega_0 n}}{j2}\right) + j2\left(\frac{e^{j2\Omega_0 n} - e^{-j2\Omega_0 n}}{j2}\right)$$

$$x[n] = 1 + j2\sin(j\Omega_0 n) + j2\sin(j2\Omega_0 n)$$

$$\boxed{x[n] = 1 + j2\sin\left(j\frac{2\pi}{5}n\right) + j2\sin\left(j\frac{4\pi}{5}n\right)}$$

3.57(d) Determine the time-domain signal corresponding to each of the following frequency-domain representations:

(d)

4/6

Frequency domain, continuous, non-periodic $\rightarrow$ FT

FT: $x(t) = \frac{1}{2\pi}\int_{-\infty}^{\infty} X(j\omega)\, e^{j\omega t}\, d\omega$

$$X(j\omega) = \begin{cases} -e^{-\omega}, & -2<\omega<0 \\ e^{-\omega}, & 0<\omega<2 \\ 0, & \text{otherwise} \end{cases}$$

$$x(t) = \frac{1}{2\pi}\left[\int_{-2}^{0} -e^{-\omega} e^{j\omega t}\, d\omega + \int_{0}^{2} e^{-\omega} e^{j\omega t}\, d\omega\right] = \frac{1}{2\pi}\left[\int_{-2}^{0} -e^{\omega(-1+jt)}\, d\omega + \int_{0}^{2} e^{\omega(-1+jt)}\, d\omega\right]$$

$$= \frac{1}{2\pi}\left[\frac{-e^{\omega(-1+jt)}}{-1+jt}\Big|_{-2}^{0} + \frac{e^{\omega(-1+jt)}}{-1+jt}\Big|_{0}^{2}\right]$$

$$= \frac{1}{2\pi(-1+jt)}\left[-e^{\omega(-1+jt)}\Big|_{-2}^{0} + e^{\omega(-1+jt)}\Big|_{0}^{2}\right]$$

$$= \frac{1}{2\pi(-1+jt)}\left[\cancel{-1} + (+e^{(-2)(-1+jt)}) + \cancel{1} - e^{(2)(-1+jt)}\right]$$

$$= \frac{1}{2\pi(-1+jt)}\left[e^{(2-2jt)} + -e^{(-2+2jt)}\right] \qquad e^{a+jb} = e^{a}(\cos b + j\sin b)$$

$$= \frac{1}{2\pi(-1+jt)}\left[e^{2}(\cos(-2t) + j\sin(-2t)) - e^{-2}(\cos 2t + j\sin 2t)\right]$$

$$= \frac{1}{2\pi(-1+jt)}\left[e^{2}(\cos 2t - j\sin 2t) - e^{-2}(\cos 2t + j\sin 2t)\right]$$

$$\boxed{x(t) = \frac{1}{2\pi(-1+jt)}\left[(e^{2}-e^{-2})\cos 2t - j(e^{2}+e^{-2})\sin 2t\right]}$$

3.58 (b) Use the tables of transforms and properties to find the FT's of the following signals:

$\boxed{5/6}$

$$x(t) = t\, e^{-3|t-1|}$$

$$x(t) = e^{-a|t|},\ a>0 \xleftrightarrow{FT} X(j\omega) = \frac{2a}{a^2+\omega^2}$$

$$x(t)z(t) \xleftrightarrow{FT} \frac{1}{2\pi} X(j\omega) * Z(j\omega) \qquad X(j\omega) = \int_{-\infty}^{\infty} x(t) e^{-j\omega t}\, dt$$

$$x(t-t_0) \xleftrightarrow{FT} e^{-j\omega t_0} X(j\omega)$$

$$a(t) = e^{-3|t|} = \frac{2(3)}{(3)^2+\omega^2} = \underbrace{\frac{6}{9+\omega^2}}_{A(t)}; \quad a(t-1) \longleftrightarrow e^{-j\omega(1)} A(j\omega)$$

$$e^{-j\omega}\frac{6}{9+\omega^2} = \frac{6e^{-j\omega}}{9+\omega^2} = FT\{e^{-3|t-1|}\}$$

$FT\{a x(t)\} = aX(j\omega)$: Linearity

$FT\{-jt x(t)\} = \frac{d}{d\omega} X(j\omega)$: Differentiation in frequency

$$FT\{t x(t)\} = \frac{\frac{d}{d\omega}X(j\omega)}{-j} = j \cdot \frac{d}{d\omega} X(j\omega)$$

$$FT\{t e^{-3|t-1|}\} = j \cdot \frac{d}{d\omega}\left[\frac{6e^{-j\omega}}{9+\omega^2}\right] = 6j \frac{d}{d\omega}\left[\frac{e^{-j\omega}}{9+\omega^2}\right]$$

$$6j \cdot \left[\frac{(9+\omega^2)\left(\frac{e^{-j\omega}}{-j}\right) - (e^{-j\omega})(2\omega)}{(9+\omega^2)^2}\right]$$

$$\boxed{X(t) = j6e^{-j\omega}\left[\frac{j}{9+\omega^2} - \frac{2\omega}{(9+\omega^2)^2}\right]}$$

3.60(c) Use the tables of transforms and properties to find the DTFT of the following signal: $\boxed{6/6}$

$$x[n] = \cos\left(\tfrac{\pi}{4}n\right)\left(\tfrac{1}{2}\right)^n u[n-2]$$

Discrete, time, non-periodic → DTFT ✓

C.6 : $x[n] = \cos(\Omega_1 n) \overset{DTFT}{\longleftrightarrow} X(e^{j\Omega}) = \pi \sum_{k=-\infty}^{\infty}\left[\delta(\Omega-\Omega_1-k2\pi) + \delta(\Omega+\Omega_1-k2\pi)\right]$

C.3 : $x[n] = \alpha^n u[n] \overset{DTFT}{\longleftrightarrow} X(e^{j\Omega}) = \frac{1}{1-\alpha e^{-j\Omega}}$ C.7 : $x[n-n_0] \overset{DTFT}{\longleftrightarrow} e^{-j\Omega n_0} X(e^{j\Omega})$

C.7 ; $x[n]y[n] \overset{DTFT}{\longleftrightarrow} \frac{1}{2\pi}\int_{-\pi}^{\pi} X(e^{j\theta}) Y(e^{j(\Omega-\theta)})\, d\theta$

$$FT\{\underbrace{\cos(\tfrac{\pi}{4}n)}_{z[n]}\} = \pi\sum_{k=-\infty}^{\infty}\left[\delta(\Omega-\tfrac{\pi}{4}-k2\pi) + \delta(\Omega+\tfrac{\pi}{4}-k2\pi)\right] = Z(e^{j\Omega})$$

$w[n] = \left(\frac{1}{2}\right)^n u[n-2] = \left(\frac{1}{2}\right)^2\left(\frac{1}{2}\right)^{n-2} u[n-2]$; C.7 $FT\{ax[n]\} = aX(e^{j\Omega})$ (linearity)

$y[n] = \left(\frac{1}{2}\right)^{n-2} u[n-2]$

$b[n] = \frac{1}{2}^n u[n] \longrightarrow FT\{b[n]\} = \frac{1}{1-\frac{1}{2}e^{-j\Omega}} = B(e^{j\Omega})$: base for $w[n]$

$FT\{y[n]\} = e^{-j\Omega 2} B(e^{j\Omega}) = Y(e^{j\Omega}) = e^{-j(\Omega)(2)} \cdot \frac{1}{1-\frac{1}{2}e^{-j\Omega}} = \frac{e^{-j\Omega 2}}{1-\frac{1}{2}e^{-j\Omega}}$: time shift

$w[n] = \left(\frac{1}{2}\right)^2 y[n] \longrightarrow FT\{\left(\tfrac{1}{2}\right)^2 y[n]\} = \left(\tfrac{1}{2}\right)^2 \cdot Y[n] = \left(\tfrac{1}{2}\right)^2 \cdot \frac{e^{-j\Omega 2}}{1-\frac{1}{2}e^{-j\Omega}} = \frac{\left(\frac{1}{2}\right)^2 e^{-j\Omega 2}}{1-\frac{1}{2}e^{-j\Omega}} = W(e^{j\Omega})$

$$x[n] = w[n]z[n] = \frac{1}{2\pi}\int_{-\pi}^{\pi} W(e^{j\theta}) Z(e^{j(\Omega-\theta)})\, d\theta$$

$$= \frac{1}{2\pi}\int_{-\pi}^{\pi}\left[\left\{\frac{\left(\frac{1}{2}\right)^2 e^{-j\theta 2}}{1-\frac{1}{2}e^{-j\theta}}\right\} \cdot \left\{\pi \underbrace{\sum_{k=-\infty}^{\infty}}_{k=0 \text{ for 1 period}}\left[\delta(\Omega-\tfrac{\pi}{4}-k2\pi-\theta) + \delta(\Omega+\tfrac{\pi}{4}-k2\pi-\theta)\right]\right\}\right] d\theta$$

$$X(e^{j\Omega}) = \frac{1}{2\pi}\int_{-\pi}^{\pi}\left[\left[\frac{\left(\frac{1}{2}\right)^2 e^{-j\theta 2}}{1-\frac{1}{2}e^{-j\theta}}\right] \cdot \left[\pi\,\delta(\Omega-\tfrac{\pi}{4}-\theta) + \pi\,\delta(\Omega+\tfrac{\pi}{4}-\theta)\right]\right] d\theta$$

sifting property of an impulse : $f(t_0) = \int_{-\infty}^{\infty} f(t)\,\delta(t-t_0)\,dt$

dist. first term to both δ

$\delta(-x) = \delta(x)$; $\delta(\Omega \pm \frac{\pi}{4} - \theta) = \delta(-(\theta - (\Omega \pm \frac{\pi}{4}))) = \delta(\theta - (\Omega \pm \frac{\pi}{4}))$

$$\boxed{X(e^{j\Omega}) = \frac{\left(\frac{1}{2}\right)^2 e^{-j2(\Omega-\pi/4)}}{1-\frac{1}{2}e^{-j(\Omega-\pi/4)}} + \frac{\left(\frac{1}{2}\right)^2 e^{-j2(\Omega+\pi/4)}}{1-\frac{1}{2}e^{-j(\Omega+\pi/4)}}}$$

(3.112)

Evaluate the frequency response of the truncated filter in Example 3.46. You can do this in MATLAB by writing an m-file to evaluate

$$H_t(e^{j\Omega}) = \sum_{n=-M}^{M} h[n]e^{-j\Omega n}$$

for a large number (> 1000) of values of Ω in the interval $-\pi < \Omega \leq \pi$ Plot the frequency response magnitude in dB $(20log_{10}|H_t(e^{j\Omega})|)$ for the following values of M:

(a) M=4

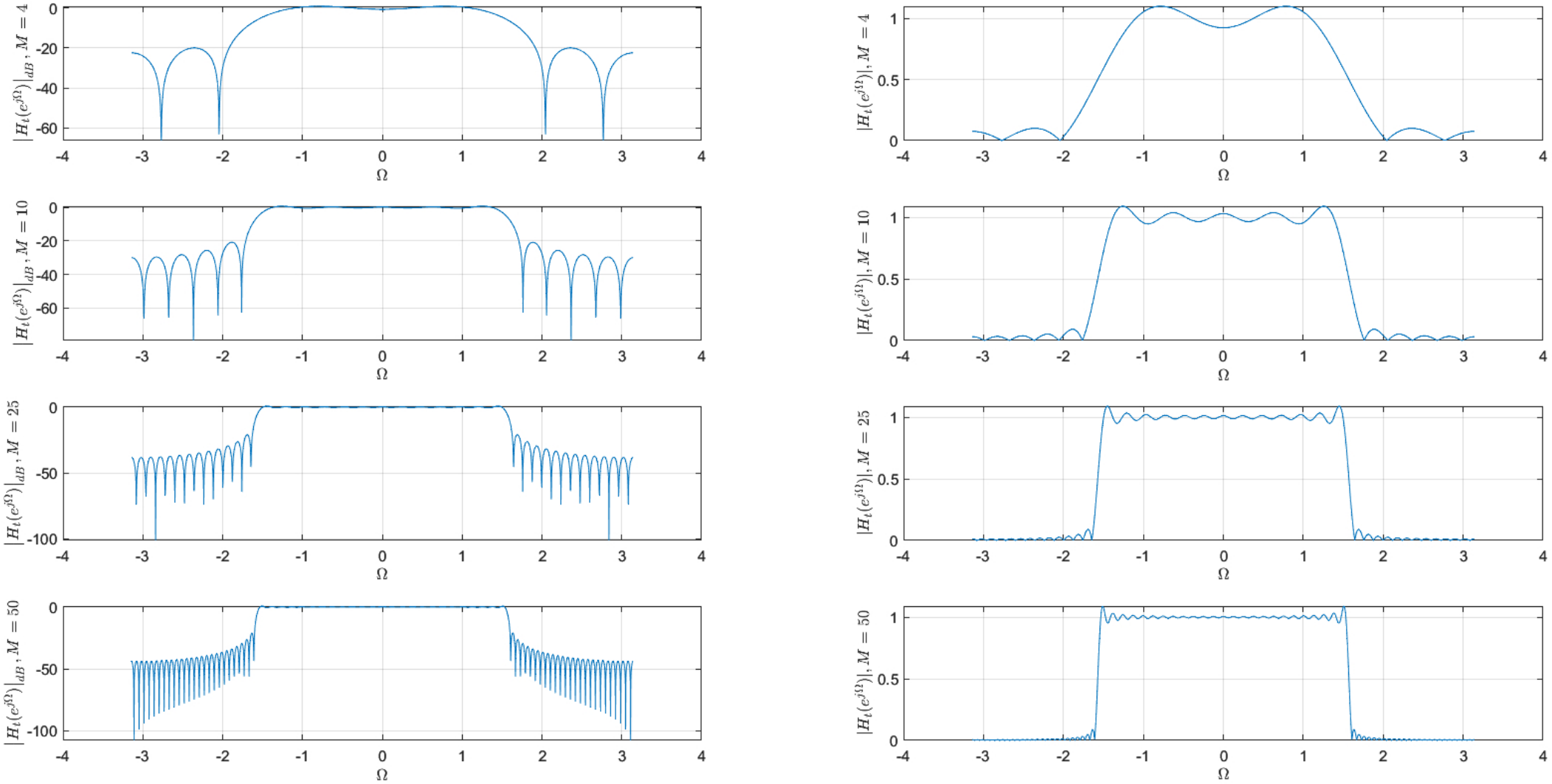
(b) M=10

(c) M=25

(d) M=50

Discuss the effect of increasing M on the accuracy with which $H_t(e^{j\Omega})$ approximates $H(e^{j\Omega})$.

```
% From "W08_PS8 Examples_wide_25to54.pdf" in Canvas
% MATLAB script for MATLAB 3.112
clear
M = [4,10,25,50];  % M = 166 makes a cool logo in dB plot
for k = 1:4
    Ht = 0;
    Omega = -pi:pi/1000:pi;
    for n = -M(k):M(k)
        if n ~= 0
            hn = 1/pi/n*sin(pi*n/2);
        else
            hn = 1/2;
        end
        Ht = Ht + hn*exp(-1j*Omega*n);
    end
    Htmag = abs(Ht);
    HtdB = 20*log10(abs(Ht));
    figure(1)
    subplot(4,2,2*k-1)
    plot(Omega, HtdB)
    grid on
    xlabel('$\Omega$', 'interpreter', 'latex')
    y_str = sprintf('%s%i%s', '$\left| H_t(e^{j\Omega}) \right|_{dB}, M=', M(k), '$');
    ylabel(y_str, 'Interpreter', 'latex')

    subplot(4,2,2*k)
    plot(Omega, Htmag)
    grid on
    xlabel('$\Omega$', 'interpreter', 'latex')
    y_str = sprintf('%s%i%s', '$|H_t(e^{j\Omega})|, M=', M(k), '$');
    ylabel(y_str, 'Interpreter', 'latex')
end
sgtitle('\underline{Frequency Response of the Truncated Filter: Seth Ricks}', 'Interpreter', 'latex'); % Ov
```

Frequency Response of the Truncated Filter: Seth Ricks